§2.5

P1. solution:

$$A^{-1} = -\frac{1}{12}\begin{bmatrix} 0 & -3 \\ -4 & 0 \end{bmatrix}$$

$$B^{-1} = \frac{1}{4}\begin{bmatrix} -2 & 4 \\ 0 & -2 \end{bmatrix}$$

$$C^{-1} = \begin{bmatrix} 7 & -4 \\ -5 & 3 \end{bmatrix}$$

P7. solution:

(a) $A = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$ $\quad a_1 + a_2 = a_3$

If $AX = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ for some X. $\Rightarrow$ $a_1 \cdot x = 1$, $a_2 \cdot x = 0$, $a_3 \cdot x = 0$

$$\Rightarrow 1 = (a_1 + a_2)x = a_3 x = 0$$

Contradiction!

(b) $AX = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$

$a_1 X = b_1$, $a_2 \cdot X = b_2$, $a_3 \cdot X = b_3$ $\Rightarrow$ $b_1 + b_2 = (a_1 + a_2) \cdot X = a_3 \cdot X = b_3$

So that for $AX = b$ to have a solution, it must hold $b_1 + b_2 = b_3$

(c) $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \rightarrow \begin{bmatrix} * & * & * \\ * & * & * \\ 0 & 0 & 0 \end{bmatrix}$

P12. Solution:

$$C = AB$$

$$A^{-1}C = A^{-1}AB = B$$

$$A^{-1}CC^{-1} = BC^{-1}$$

$$\Rightarrow A^{-1} = BC^{-1}$$

P16. Solution:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = (ad-bc)I$$

$$\Rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

P22. Solution:

(a) $$[A\ I] = \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{array}\right] = [I\ A^{-1}]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} 7 & -3 \\ -2 & 1 \end{bmatrix}$$

(b) $$[A\ I] = \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 3 & 9 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & -3 & -3 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & -3 & \frac{4}{3} \\ 0 & -3 & -3 & 1 \end{array}\right] \rightarrow \left[\begin{array}{cc|cc} 1 & 0 & -3 & \frac{4}{3} \\ 0 & 1 & 1 & -\frac{1}{3} \end{array}\right] = [I\ A^{-1}]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} -3 & \frac{4}{3} \\ 1 & -\frac{1}{3} \end{bmatrix}$$

P25. Solution:

(a) $$[A\ I] = \left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & \frac{3}{2} & -\frac{1}{2} & 0 & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & \frac{1}{2} & -\frac{1}{2} & 1 & 0 \\ 0 & 0 & \frac{4}{3} & -\frac{1}{3} & -\frac{1}{3} & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 2 & 1 & 0 & \frac{5}{4} & \frac{1}{4} & -\frac{3}{4} \\ 0 & \frac{3}{2} & 0 & -\frac{3}{8} & \frac{9}{8} & -\frac{3}{8} \\ 0 & 0 & \frac{4}{3} & -\frac{1}{3} & -\frac{1}{3} & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 2 & 0 & 0 & \frac{3}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{3}{2} & 0 & -\frac{3}{8} & \frac{9}{8} & -\frac{3}{8} \\ 0 & 0 & \frac{4}{3} & -\frac{1}{3} & -\frac{1}{3} & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & 1 & 0 & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ 0 & 0 & 1 & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{array}\right] = [I\ A^{-1}]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \end{bmatrix}$$

(b) $[B\ I] = \left[\begin{array}{ccc:ccc} 2 & -1 & -1 & 1 & 0 & 0 \\ -1 & 2 & -1 & 0 & 1 & 0 \\ -1 & -1 & 2 & 0 & 0 & 1 \end{array}\right]$

$\rightarrow \left[\begin{array}{ccc:ccc} 2 & -1 & -1 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & -\frac{3}{2} & \frac{1}{2} & 1 & 0 \\ 0 & -\frac{3}{2} & \frac{3}{2} & \frac{1}{2} & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc:ccc} 2 & -1 & -1 & 1 & 0 & 0 \\ 0 & \frac{3}{2} & -\frac{3}{2} & \frac{1}{2} & 1 & 0 \\ \underbrace{0 \quad 0 \quad 0}_{\text{no pivots}} & & & 1 & 1 & 1 \end{array}\right]$

$\Rightarrow B$ is not invertible

P35. Solution:

i) $\begin{bmatrix} I & 0 \\ C & I \end{bmatrix}^{-1} = \begin{bmatrix} I & 0 \\ -C & I \end{bmatrix}$

ii) $\left[\begin{array}{cc:cc} A & 0 & I & 0 \\ C & D & 0 & I \end{array}\right] \xrightarrow{CA^{-1}} \left[\begin{array}{cc:cc} A & 0 & I & 0 \\ 0 & D & -CA^{-1} & I \end{array}\right]$

$\rightarrow \left[\begin{array}{cc:cc} I & 0 & A^{-1} & 0 \\ 0 & I & -D^{-1}CA^{-1} & D^{-1} \end{array}\right]$

$\Rightarrow \begin{bmatrix} A & 0 \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} & 0 \\ -D^{-1}CA^{-1} & D^{-1} \end{bmatrix}$

iii) $\left[\begin{array}{cc:cc} 0 & I & I & 0 \\ I & D & 0 & I \end{array}\right] \rightarrow \left[\begin{array}{cc:cc} I & D & 0 & I \\ 0 & I & I & 0 \end{array}\right]$

$\xrightarrow{D^{-1}} \left[\begin{array}{cc:cc} I & 0 & -D & I \\ 0 & I & I & 0 \end{array}\right]$

$\Rightarrow \begin{bmatrix} 0 & I \\ I & D \end{bmatrix}^{-1} = \begin{bmatrix} -D & I \\ I & 0 \end{bmatrix}$

§2.6.

P1. Solution:

$l_{21} = 1$

$L = \begin{bmatrix} 1 & \\ 1 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} x = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$

$LUx = Lc$

P2 Solution:

$l_{31} = 1$

$l_{32} = 2$

P6. solution:

$E_{21} = \begin{bmatrix} 1 & & \\ -2 & 1 & \\ & & 1 \end{bmatrix}$

$E_{32} = \begin{bmatrix} 1 & & \\ & 1 & \\ & -2 & 1 \end{bmatrix}$

$L = \begin{bmatrix} 1 & & \\ 2 & 1 & \\ 0 & 2 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & -6 \end{bmatrix}$

P11. solution:

$A = \begin{bmatrix} 2 & 4 & 8 \\ 0 & 3 & 9 \\ 0 & 0 & 7 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} 2 & & \\ & 3 & \\ & & 7 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ & 1 & 3 \\ & & 1 \end{bmatrix}$

$L = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix} \quad D = \begin{bmatrix} 2 & 4 & 8 \\ 0 & 3 & 9 \\ 0 & 0 & 7 \end{bmatrix}$ in LU

$L = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 1 \end{bmatrix}, D = \begin{bmatrix} 2 & & \\ & 3 & \\ & & 7 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 & 4 \\ & 1 & 3 \\ & & 1 \end{bmatrix}$ in LDU

P12. Solution:

i) $A = \begin{bmatrix} 2 & 4 \\ 4 & 11 \end{bmatrix} \xrightarrow{l_{21}=2} \begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix}$

$\Rightarrow A = \begin{bmatrix} 1 & \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ & 3 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \\ 2 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & \\ & 3 \end{bmatrix}}_{D} \underbrace{\begin{bmatrix} 1 & 2 \\ & 1 \end{bmatrix}}_{U}$

$L^T = U$!

ii) $B = \begin{bmatrix} 1 & 4 & 0 \\ 4 & 12 & 4 \\ 0 & 4 & 0 \end{bmatrix} \xrightarrow{l_{21}=4} \begin{bmatrix} 1 & 4 & 0 \\ 0 & -4 & 4 \\ 0 & 4 & 0 \end{bmatrix}$

$\xrightarrow{l_{32}=-1} \begin{bmatrix} 1 & 4 & 0 \\ 0 & -4 & 4 \\ 0 & 0 & 4 \end{bmatrix}$

$\Rightarrow A = \begin{bmatrix} 1 & & \\ 4 & 1 & \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 0 \\ & -4 & 4 \\ & & 4 \end{bmatrix}$

$= \underbrace{\begin{bmatrix} 1 & & \\ 4 & 1 & \\ 0 & -1 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & & \\ & -4 & \\ & & 4 \end{bmatrix}}_{D} \underbrace{\begin{bmatrix} 1 & 4 & 0 \\ & 1 & -1 \\ & & 1 \end{bmatrix}}_{U} \quad L^T = U$!

P16. Solution:

$Lc = b$

$$\begin{bmatrix} 1 & & \\ 1 & 1 & \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$\Rightarrow c_1 = 4, \quad c_2 = 1, \quad c_3 = 1$

$Ux = c$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$$

$\Rightarrow x_3 = 1, \ x_2 = 0, \ x_1 = 3$

$$A = LU = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$